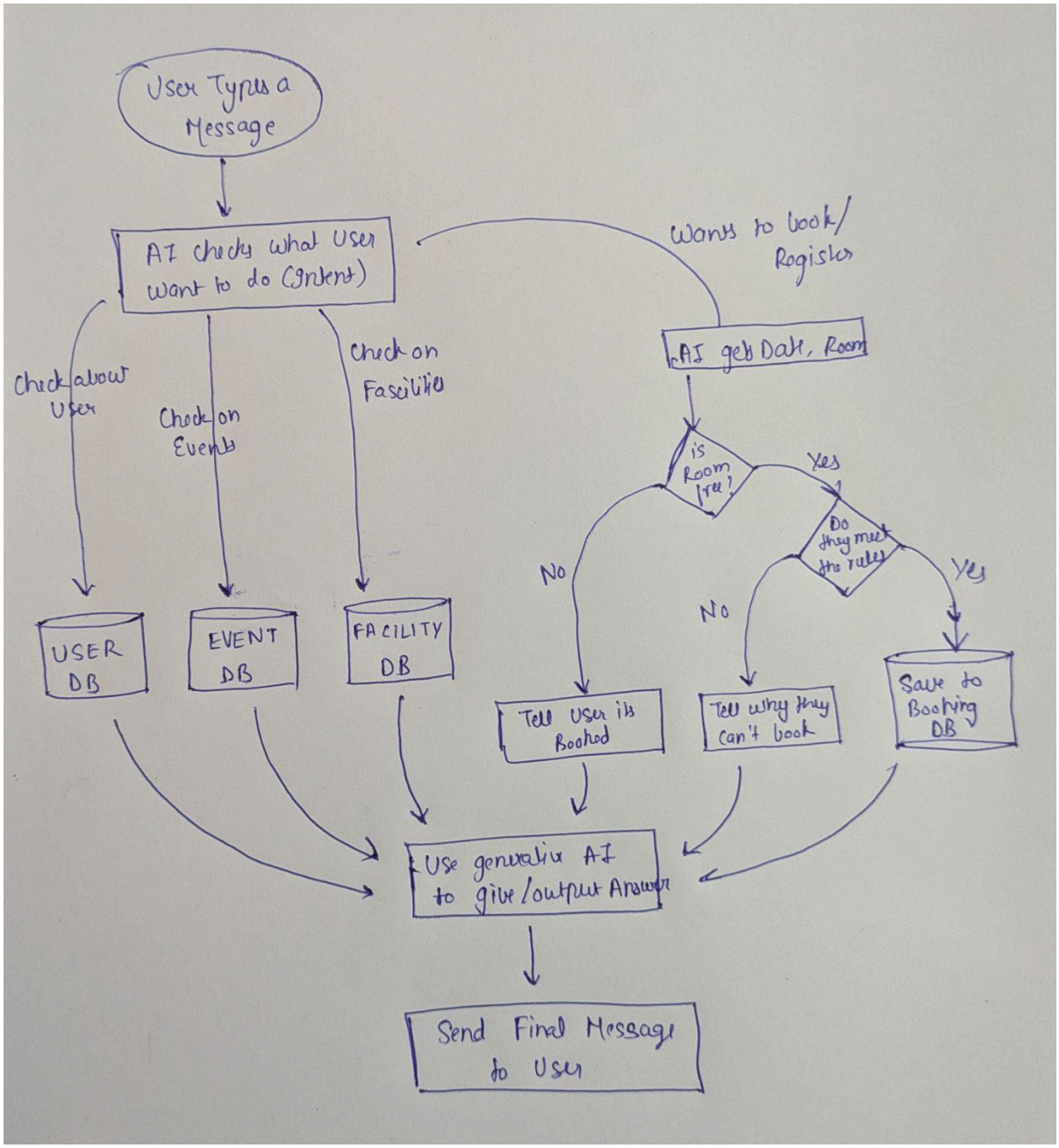
Campus AI Chatbot - System Architecture

1. System Flow Diagram



2. Architecture Explanation

This AI chatbot is designed with a simple, step-by-step flow to help students on campus. When a user sends a message.

AI first analyzes the intent to figure out their goal such as checking event times, asking about campus rules, or booking a lab. If they just need information, the system quickly searches the correct database (like the Campus Events DB or User DB or the Facilities Rules DB) and uses the AI to draft a clear, helpful answer. This keeps basic questions fast and easy to handle.

If the user wants to book a room, the process includes a few extra safety steps. The AI extracts the requested date and room, then checks the database to see if the room is actually free and if the student is allowed to use it. If everything looks good, the chatbot saves the transaction to the database.

3. Tech Stack & Libraries

- **Programming Language: Python** Python is the most beginner-friendly language and is the standard choice for building basic AI and data projects.
- **AI Integration: OpenAI API (openai Python library)** Instead of trying to build and train a complex machine learning model from scratch, using the OpenAI API allows the chatbot to easily understand the user's intent, extract booking details, and generate natural-sounding text responses.
- **Web Framework: Flask** Flask is a lightweight and simple Python framework. It is perfect for connecting a basic chat interface to the backend logic.
- **Databases: SQLite & MongoDB** SQLite is used for the booking ledger because it is built right into Python and is perfect for handling structured relational data safely. MongoDB is used for the events database because it stores data in a flexible format.